

Vincent Quirion

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EDUCATION

Mila - University of Montreal

Master's Degree, Computer Science - GPA: 4.0

Montreal, Canada

September 2024 – Anytime

University of Montreal

Bachelor's Degree, Honors Computer Science – GPA: 4.0

Montreal, Canada

Sept. 2021 – April 2024

WORK EXPERIENCE

Machine Learning Researcher | collaboration with Google DeepMind

June 2024 – Present

Mila - Montreal Institute for Learning Algorithms - University of Montreal

Montreal, Canada

- Fundamental research on machine unlearning in collaboration with Google DeepMind researchers Karolina Dziugaite, Eleni Triantafillou, and Ioannis Mitliagkas. [Paper to submitted to ICML 2026](#).
- Studied the evolution of the discretized Langevin stochastic differential equation (noisy SGD) to prove statistical indistinguishability guarantees and generalization bounds for the Langevin Unlearning algorithm.
- Introduced a framework for incorporating auxiliary public domain data to improve the unlearning-utility tradeoff of Langevin Unlearning, and derived extended bounds for this regime.
- Ran large-scale experiments that highlighted the benefits of this novel framework for Langevin Unlearning.

Machine Learning Researcher | with Prof. Yoshua Bengio

September 2022 – May 2024

Mila - Montreal Institute for Learning Algorithms - University of Montreal

Montreal, Canada

- Led the development of [an embarrassingly parallel method](#) to better scale the training of generative models on multi-GPU hardware.
- Implemented it in efficient distributed PyTorch code and ran experiments on multi-node hardware configurations of the Mila cluster.
- Collaborated with Recursion Pharmaceuticals researchers to test the method on their industry-grade molecule generation environment.

Quantum Information Science Researcher | with Prof. Gilles Brassard

May 2023 – August 2023

Laboratory for Theoretical and Quantum Informatics - University of Montreal

Montreal, Canada

- Utilized the [NPA hierarchy](#) to approximate the best strategies for non-local games by solving semi-definite programs. Work presented in my collaborator's [thesis](#) about entanglement-assisted communication complexity.
- Wrote high performance distributed Julia code to efficiently solve these SDPs at scale on a compute cluster.

Software Engineer

May 2021 – September 2022

Secure AI Labs - Massachusetts Institute of Technology (MIT) spin-off

Cambridge, Massachusetts

- Implemented numerous features of their proprietary federated learning and differential privacy library.
- Wore many other software engineer hats to build the MVP that allowed SAIL to raise a [\\$5 million seed round](#).

RELEVANT COURSEWORK

Adversarial Learning (IFT6164) | Mila - University of Montreal | Grade: A+

Winter 2025

- Material: adversarial examples, adversarial learning, generative adversarial networks, Wasserstein generative adversarial networks, optimization & differentiable games, extragradient method, spectral analysis, stability and equilibrium, multi agent RL, evaluation and learning in multi-agent systems.

Probabilistic Graphical Models (IFT6269) | Mila - University of Montreal | Grade: A+

Fall 2024

- Material: probability concepts, maximum likelihood estimation, linear regression, logistic regression, Fisher discriminant, K-means, expectation-maximization algorithm, Gaussian mixtures, directed and undirected graphical models, exponential family, information theory, Gaussian networks, factor analysis, sum-product algorithm, hidden markov models, junction tree, approximate inference & sampling, variational methods, estimation of parameters in graphical models, Bayesian methods, model selection.